

LABORATORY HAZARD ASSESSMENT CHECKLIST

This list is designed to help you answer five critical questions prior to beginning work:

1. What are the hazards?
2. What is the worst thing that could happen?
3. What can be done to prevent this from happening?
4. What can be done to protect from these hazards?
5. What should be done if something goes wrong?

Pre-Operational Planning

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| ☐ Toxicity | What is the level of toxicity? What are the routes of exposure (inhalation, skin absorption, ingestion, injection) and which of these are likely under the conditions of use? What are the signs and symptoms of overexposure? |
| ☐ Flammability | Is the material flammable or explosive under the conditions of use? |
| ☐ Warning Properties | Can odor or irritation adequately warn of over-exposure before it becomes dangerous? |
| ☐ Laboratory Equipment | Is laboratory equipment in good condition? Are machine guards or interlocks in place and functioning? |
| ☐ Storage Precautions | Does the material need isolated storage, refrigeration or other special conditions for storage? |
| ☐ Incompatible Materials | Should certain materials be segregated (e.g., flammables and oxidizers)? |
| ☐ Reagent Stability | Should materials be dated for disposal (e.g., ethers)? Should materials be kept refrigerated to prolong shelf life? |
| ☐ Protective Clothing | Is a lab coat, apron, or clothing made of resistant material needed or is a standard lab coat adequate? |
| ☐ Gloves | What glove material is needed? Is the right type, thickness, glove length and size available for the materials being handled? |
| ☐ Eye Protection | What type of eye protection is needed (e.g., safety glasses for impact, chemical splash goggles for chemicals)? Is a face shield needed in combination with the goggles? |
| ☐ Heat Sources | Is heating needed? Is there an alternative to open flames? Are heating mantles in good condition? |
| ☐ Electrical Equipment | Is equipment it grounded and bonded properly? Are electrical cords insulated? Is ground fault circuit interruption (GFCI) needed? |

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| ☐ Vacuum/Pressure Systems | Have connections been leak tested, hydrostatically tested, properly vented, and traps installed when necessary? |
| ☐ Ventilation/Containment | Does the work need to be done in a chemical hood, ventilated cabinet or a glove box to provide the needed level of containment? |

Experimental Scale & Design

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| ☐ Quantity | Are there ways to minimize the amount of materials used without affecting results (e.g. microscale)? |
| ☐ Ambient Conditions | Are special conditions necessary to carry out the reaction (e.g., cold room or dry box)? |
| ☐ Time Constraints | Can the experiment be completed while lab workers are present? If not, can the experiment be safely run unattended or overnight? |

Spill/Emergency Planning

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| ☐ Lab Personnel | Are others in the laboratory aware of what you are doing? |
| ☐ Fire Extinguishers | Are special types required; are you aware of their location and proper use (e.g., Class D for metals)? |
| ☐ Emergency Response | Do you have a response planned in the event of a spill; would evacuation be necessary? |
| ☐ Spill Cleanup | Are materials on hand to absorb/neutralize; is the needed protective equipment on hand and have you been trained on its use? |
| ☐ Safety Shower/Eyewash Fountain | Are you aware of the locations and methods of operation? |

Waste Disposal

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| ☐ Method | Is there an approved method for disposal of waste generated by the experiment or procedure? |
| ☐ Labeling | Are waste containers clearly, indelibly and accurately labeled as to the contents? |
| ☐ Segregation | Are incompatible wastes kept segregated? |
| ☐ Containers | Are suitable containers with adequate closures available? |
| ☐ Recycling | Is it feasible to safely recover/recycle used chemical? |